THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 126	Find the breadth first.
2	**(c)** 144	Area becomes 4 times.
3	**(d)** 4800	Area $\times$ rate.
4	**(d)** 9	Perimeter of square $= 4\times$ side.
5	**(d)** 100	Side of square $=$ perimeter $\div 4$.
6	**(a)** 256	Inner side $= a - 2 \times 2$.
7	**(a)** 9	Height $= 2\times$ area $\div$ base.
8	**(c)** 54	Half of the rectangle's area.
9	**(c)** 154	Area $= l \times b$.
10	**(d)** 275	Find the breadth first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$l + b = 35$, so breadth $= 11$ cm (1); Area $= 264$ cm^2 (1).	1+1
12	Area $= 96$ m^2 (1); Cost $= 96 \times 75 =$ ₹7200 (1).	1+1
13	New side 12 cm; New area 144 cm^2; old area 100 cm^2 (1); Increase $= 44$ cm^2 (1).	1+1
14	$l + b = 47$, so breadth $= 15$ cm (1); Area $= 480$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inner lawn: 26 m by 16 m (1); Inner area 416 m^2 (1); Path area $= 600 - 416 = 184$ m^2 (1).	1+1+1
16	First area $= 80$ cm^2 (1); Second area $= 144$ cm^2 (1); Larger: second, difference $= 64$ cm^2 (1).	1+1+1
17	Along the length: 5 squares; along the width: 3 squares (1); Squares $= 15$ (1); Left over area $= 250$ cm^2 (1).	1+1+1
	OR: Inner lawn: 34 m by 19 m (1); Inner area 646 m^2 (1); Path area $= 1000 - 646 = 354$ m^2 (1).	1+1+1
18	First area $= 108$ cm^2 (1); Second area $= 140$ cm^2 (1); Larger: second, difference $= 32$ cm^2 (1).	1+1+1
	OR: Along the length: 8 squares; along the width: 5 squares (1); Squares $= 40$ (1); Left over area $= 320$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hall in cm: 1000 by 800 (1); Tiles along each side: 20 and 16 (1); Number of tiles $= 320$ (1); Cost $=$ ₹12800 (2).	1+1+1+2
	OR: Area of big square $= 400$ cm^2 (1); Area of four small squares $= 4 \times 9 = 36$ cm^2 (1); Remaining area $= 364$ cm^2 (1); Perimeter stays the same as the original $= 4 \times 20 = 80$ cm (1); Cutting corner squares does not change the perimeter (1).	1+1+1+1+1
20	Hall in cm: 1200 by 900 (1); Tiles along each side: 30 and 22 (1); Number of tiles $= 660$ (1); Cost $=$ ₹16500 (2).	1+1+1+2
	OR: Area of big square $= 576$ cm^2 (1); Area of four small squares $= 4 \times 16 = 64$ cm^2 (1); Remaining area $= 512$ cm^2 (1); Perimeter stays the same as the original $= 4 \times 24 = 96$ cm (1); Cutting corner squares does not change the perimeter (1).	1+1+1+1+1

Mistakes to avoid:

- Using area units (cm^2) for perimeter or linear units (cm) for area.
- Forgetting the $\frac{1}{2}$ in the area of a triangle.
- Using the sloping side instead of the perpendicular height in the triangle area.
- Writing $P = l + b$ for a rectangle (it needs $2(l + b)$) or $A = 2(l + b)$.
- Mixing units: multiplying metres by centimetres without converting first.
- Using area when the question is about fencing (perimeter) and the reverse for paving or tiling.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 324	Area becomes 4 times.
2	**(a)** 9000	Area $\times$ rate.
3	**(d)** 12	Perimeter of square $= 4\times$ side.
4	**(b)** 196	Side of square $=$ perimeter $\div 4$.
5	**(a)** 400	Inner side $= a - 2 \times 2$.
6	**(a)** 11	Height $= 2\times$ area $\div$ base.
7	**(c)** 150	Half of the rectangle's area.
8	**(a)** 204	Area $= l \times b$.
9	**(d)** 420	Find the breadth first.
10	**(d)** 576	Area becomes 4 times.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $= 150$ m^2 (1); Cost $= 150 \times 60 =$ ₹9000 (1).	1+1
12	New side 15 cm; New area 225 cm^2; old area 144 cm^2 (1); Increase $= 81$ cm^2 (1).	1+1
13	$l + b = 58$, so breadth $= 18$ cm (1); Area $= 720$ cm^2 (1).	1+1
14	Area $= 280$ m^2 (1); Cost $= 280 \times 45 =$ ₹12600 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inner lawn: 32 m by 20 m (1); Inner area 640 m^2 (1); Path area $= 864 - 640 = 224$ m^2 (1).	1+1+1
16	First area $= 192$ cm^2 (1); Second area $= 252$ cm^2 (1); Larger: second, difference $= 60$ cm^2 (1).	1+1+1
17	Along the length: 6 squares; along the width: 4 squares (1); Squares $= 24$ (1); Left over area $= 444$ cm^2 (1).	1+1+1
	OR: Inner lawn: 42 m by 22 m (1); Inner area 924 m^2 (1); Path area $= 1500 - 924 = 576$ m^2 (1).	1+1+1
18	First area $= 300$ cm^2 (1); Second area $= 396$ cm^2 (1); Larger: second, difference $= 96$ cm^2 (1).	1+1+1
	OR: Along the length: 6 squares; along the width: 4 squares (1); Squares $= 24$ (1); Left over area $= 180$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hall in cm: 1500 by 1000 (1); Tiles along each side: 30 and 20 (1); Number of tiles $= 600$ (1); Cost $=$ ₹21000 (2).	1+1+1+2
	OR: Area of big square $= 900$ cm^2 (1); Area of four small squares $= 4 \times 25 = 100$ cm^2 (1); Remaining area $= 800$ cm^2 (1); Perimeter stays the same as the original $= 4 \times 30 = 120$ cm (1); Cutting corner squares does not change the perimeter (1).	1+1+1+1+1
20	Hall in cm: 900 by 600 (1); Tiles along each side: 30 and 20 (1); Number of tiles $= 600$ (1); Cost $=$ ₹9000 (2).	1+1+1+2
	OR: Area of big square $= 1296$ cm^2 (1); Area of four small squares $= 4 \times 36 = 144$ cm^2 (1); Remaining area $= 1152$ cm^2 (1); Perimeter stays the same as the original $= 4 \times 36 = 144$ cm (1); Cutting corner squares does not change the perimeter (1).	1+1+1+1+1

Mistakes to avoid:

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- Forgetting the $\frac{1}{2}$ in the area of a triangle.
- Using the sloping side instead of the perpendicular height in the triangle area.
- Writing $P = l + b$ for a rectangle (it needs $2(l + b)$) or $A = 2(l + b)$.
- Mixing units: multiplying metres by centimetres without converting first.
- Using area when the question is about fencing (perimeter) and the reverse for paving or tiling.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Perimeter and Area** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7560	Area × rate.
2	**(b)** 15	Perimeter of square = 4× side.
3	**(d)** 324	Side of square = perimeter ÷4.
4	**(b)** 676	Inner side = $a - 2 \times 2$.
5	**(d)** 7	Height = 2× area ÷ base.
6	**(a)** 198	Find the breadth first.
7	**(c)** 900	Area becomes 4 times.
8	**(b)** 7200	Area × rate.
9	**(b)** 18	Perimeter of square = 4× side.
10	**(c)** 361	Side of square = perimeter ÷4.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New side 19 cm; New area 361 cm^2; old area 225 cm^2 (1); Increase = 136 cm^2 (1).	1+1
12	$l + b = 56$, so breadth = 20 cm (1); Area = 720 cm^2 (1).	1+1
13	Area = 162 m^2 (1); Cost = 162×80 = ₹12960 (1).	1+1
14	New side 25 cm; New area 625 cm^2; old area 400 cm^2 (1); Increase = 225 cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inner lawn: 39 m by 22 m (1); Inner area 858 m^2 (1); Path area = $1260 - 858 = 402$ m^2 (1).	1+1+1
16	First area = 80 cm^2 (1); Second area = 110 cm^2 (1); Larger: second, difference = 30 cm^2 (1).	1+1+1
17	Along the length: 6 squares; along the width: 4 squares (1); Squares = 24 (1); Left over area = 1600 cm^2 (1).	1+1+1
	OR: Inner lawn: 50 m by 30 m (1); Inner area 1500 m^2 (1); Path area = $2400 - 1500 = 900$ m^2 (1).	1+1+1
18	First area = 132 cm^2 (1); Second area = 195 cm^2 (1); Larger: second, difference = 63 cm^2 (1).	1+1+1
	OR: Along the length: 9 squares; along the width: 6 squares (1); Squares = 54 (1); Left over area = 246 cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hall in cm: 1600 by 1200 (1); Tiles along each side: 20 and 15 (1); Number of tiles = 300 (1); Cost = ₹27000 (2).	1+1+1+2
	OR: Area of big square = 1600 cm^2 (1); Area of four small squares = $4 \times 49 = 196$ cm^2 (1); Remaining area = 1404 cm^2 (1); Perimeter stays the same as the original = $4 \times 40 = 160$ cm (1); Cutting corner squares does not change the perimeter (1).	1+1+1+1+1
20	Hall in cm: 2000 by 1000 (1); Tiles along each side: 40 and 20 (1); Number of tiles = 800 (1); Cost = ₹36000 (2).	1+1+1+2
	OR: Area of big square = 784 cm^2 (1); Area of four small squares = $4 \times 16 = 64$ cm^2 (1); Remaining area = 720 cm^2 (1); Perimeter stays the same as the original = $4 \times 28 = 112$ cm (1); Cutting corner squares does not change the perimeter (1).	1+1+1+1+1

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